RetinaLake: A Unified Data Lake Framework for Harmonizing Heterogeneous Retinal Datasets

Anirudh H. A.[1,2]*, Pramath G. H.[2], Chandra Sekhar Seelamantula[2]

[1]Department of Computer Science and Engineering, PES University, Bengaluru, India
[2]Department of Electrical Engineering, Indian Institute of Science, Bengaluru, India
*anirudhamar04@gmail.com

Purpose: The development of robust ophthalmic AI models is bottlenecked by heterogeneous public data. Some of the challenges faced when unifying the datasets are that the same disease label is encoded on three different grading scales, segmentation masks arrive in disparate formats, image quality varies wildly, and almost every dataset uses its own folder layout. The first practical issue in implementing large-scale ophthalmic AI models is therefore infrastructural. We introduce RetinaLake, a unified data lake framework that standardizes any public or private retinal dataset into **ML-ready exports in under 25 lines of code.** The code and usage examples are available online [Link].

Materials and Methods: We implemented RetinaLake on a PostgreSQL schema where primary keys are deterministic UUID v5 values derived from identifying fields, making re-ingestion a safe no operation (NOP). Images are fingerprinted using exact-byte, decoded-pixel, and perceptual hashes for content-aware deduplication. An asynchronous ingestion framework processes raw data through generic format adapters and task-specific annotation handlers while automatically propagating provenance metadata. Spatial annotations are normalized by a central mask converter supporting masks, contours, polygons, attention maps, and optical-coherence tomography (OCT) layer boundaries. Disease grades are harmonized through a unified scale mapping. A caption engine resolves structured annotations through a curated ophthalmic knowledge base, applying normalization and contradiction-suppression rules to generate clinically grounded natural-language captions based on metadata and labels for each image. Finally, a typed export layer streams filtered datasets directly into Parquet for downstream AI pipelines.

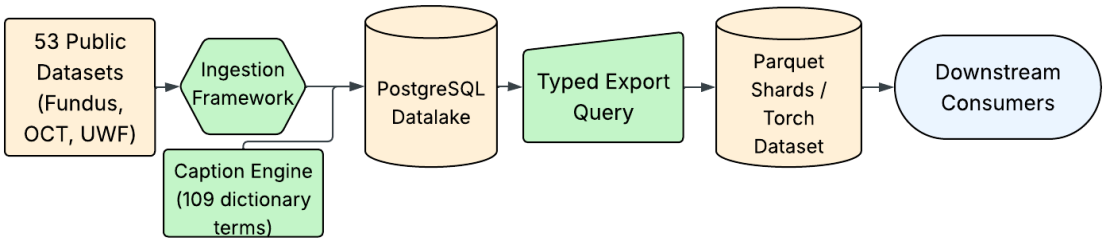


Figure 1: The RetinaLake framework architecture. The data-and-infrastructure layer unifies fifty three public datasets into a PostgreSQL data lake, enriches every image with quality and ROI metadata, and generates natural language captions.

Results: The **RetinaLake** framework unified **53** publicly available retinal **datasets**, exposing a total of **310,811 images**, and **4,037,566 annotations**. The typed query export achieved high throughput, reaching **11,618 rows per second** during benchmarking. The automated caption engine produced natural language clinical captions for images, achieving a **99.6% coverage fully consistent with their source labels**.

Conclusions: RetinaLake resolves the infrastructural data bottlenecks that hinder large-scale oculomics research. By abstracting diverse dataset integration into a simple, low-code interface, the framework provides a highly queryable, provenance-tracked, and typed view over **53** datasets. This data infrastructure provides a critical foundation that accelerates the development, training, and evaluation of any ophthalmic AI model.

Medical/clinical significance: Preventing vision loss due to diseases such as diabetic retinopathy and glaucoma are global health priorities. Harmonizing siloed clinical datasets across institutions allows researchers to combat dataset bias and build robust, generalizable diagnostic pipelines. An accessible framework for standardizing these datasets accelerates the deployment of automated clinical decision-support tools.

Prospect of application: RetinaLake's modular architecture guarantees that proprietary hospital archives and future public datasets can be exported in less than 25 lines of code. This dramatically lowers the technical barrier for clinical research groups to integrate their internal data into global training pipelines. RetinaLake has been used as an internal tool to build ChaksuAI[1], engineered for frontline ophthalmic screening at scale.

[1] Funding support from ZEISS India and Kotak-IISc AI/ML centre is gratefully acknowledged.